

Tommy Giroux

M.Sc. Candidate in Mathematics, McGill University
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EDUCATION

M.Sc. in Mathematics

2026–present

McGill University, Montréal, Québec

- Expected completion: April 2027; supervisor: Prof. John A. Toth.
- Research area: spectral geometry and semiclassical analysis of high-frequency eigenfunctions associated with completely integrable systems.

B.Sc. in Mathematics, Co-op Program

2022–2025

Université de Sherbrooke, Sherbrooke, Québec

RESEARCH INTERESTS

Spectral geometry; semiclassical and microlocal analysis; Hamiltonian dynamics; geometric analysis.

RESEARCH EXPERIENCE

M.Sc. Researcher in Mathematics

2026–present

McGill University

- Study high-frequency eigenfunctions associated with completely integrable systems, using spherical harmonics Y_k^m on S^2 as an explicit model.
- Derive allowed- and forbidden-region asymptotics through integral representations, and stationary phase, including exponential decay outside the classically allowed region.
- Identify weak limiting measures from Hamiltonian dynamics as pushforwards of normalized Liouville measure on invariant Lagrangian tori.

Research Assistant in Mathematics (Internship)

Summer 2025

Université de Sherbrooke

- Investigated infinitesimal deformations associated with hyperbolic surfaces and related questions concerning decorated hyperbolic polygons.

Research Assistant in Mathematics (Internship)

Fall 2024

Université de Sherbrooke

- Studied algebra, hyperbolic geometry, and number theory, with emphasis on fundamental domains for actions of Hilbert modular groups on $\mathbb{H} \times \mathbb{H}$.

Research Assistant in Mathematics (Internship)

Winter 2024

Université de Sherbrooke

- Investigated dynamical systems and ergodic theory; this work resulted in the expository article listed below.

Mathematical Modeler (Internship)

Summer 2023

Agriculture and Agri-Food Canada

- Synthesized models from the scientific literature and helped formulate a model of the biological cycle of grey mould disease incorporating climatic and epidemiological factors.
- Contributed to a beta disease module for an expert system and collaborated on a scientific manuscript.

UNDERGRADUATE EXPOSITORY PUBLICATION

Giroux, T. “Théorie ergodique.” *Cahiers mathématiques de l’Université de Sherbrooke (CaMUS)*, an undergraduate departmental journal, vol. 8, 2026, pp. 99–123.

TEACHING EXPERIENCE

Teaching Assistant and Course Demonstrator

2023–2025

Université de Sherbrooke

- **MQG157 – Quantitative Methods in Economics and Finance:** lab instructor (Winter and Summer 2025).
- **ROP631 – Nonlinear Optimization:** course demonstrator (Summer 2025).
- **MQG222 – Applied Statistics in Management:** lab instructor and grader (Summer 2024).
- **MAT194 – Calculus I:** course demonstrator and grader (Fall 2023 and Fall 2024).

- **CQP208 – Fundamentals of Differential Calculus:** course demonstrator (Fall 2024).

AWARDS, FELLOWSHIPS, AND SCHOLARSHIPS

FRQ Master's Research Scholarship (B1) , Nature and Technologies Sector (CAD 40,000)	2026–2028
Kenneth Eade Fellowship in Math & Statistics , McGill University (CAD 2,000)	2026
Faculty of Science Graduate Supplement Award , McGill University (CAD 5,000)	2026
Graduate Excellence Award in Mathematics & Statistics , McGill University (CAD 2,250)	2026
Pierre Breton Scholarship , Faculty of Science, Université de Sherbrooke	2025
NSERC Undergraduate Student Research Award (USRA)	2023
SSDP Challenge Scholarship , Université de Sherbrooke	2023
Entrance Scholarship , Université de Sherbrooke	2022

ACADEMIC SERVICE AND OUTREACH

President, 2022 Mathematics Cohort , Université de Sherbrooke	2022–2025
– Represent students' interests, organize academic and social events, and serve as a liaison with faculty and university administration.	
Facilitator, Sherbrooke Mathematics Circle	2024–2025
– Lead activities for children involving logic, puzzles, and abstract thinking.	
Volunteer, AQJM Mathematics Games	2024–2025
– Provide participant support and assist with event logistics.	

TECHNICAL SKILLS AND LANGUAGES

Programming: Python, R, C/C++. **Typesetting:** $\text{\LaTeX}$. **Languages:** French (native), English (fluent).